

JSS ACADEMY OF TECHNICAL EDUCATION, NOIDA

DEPARTMENT OF INFORMATION TECHNOLOGY



PROJECT REPORT

ON

DIABETES PREDICTION & RISK SCORING USING MACHINE LEARNING

Submitted as Part of the Summer Internship in Data Analytics and Intelligence

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ACADEMIC YEAR 2026–2027

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our project guides, Dr. Sukhendra Singh and Dr. Prachi Chhabra, for their invaluable guidance, encouragement, and continuous support throughout the course of this project. Their expertise and constructive feedback were instrumental in shaping the direction, methodology, and quality of this work.

We are also sincerely thankful to the Department of Information Technology, JSS Academy of Technical Education, Noida, for providing us with the opportunity, resources, and infrastructure required to carry out this project as part of our Summer Internship in Data Analytics and Intelligence.

Finally, we would like to thank our families and friends for their constant encouragement and support throughout this project.

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DECLARATION

We, Anushka Gupta, Tanushi Taparia, and Yati Rastogi, hereby declare that the project report entitled “Diabetes Prediction & Risk Scoring Using Machine Learning”, submitted in partial fulfilment of the requirements for the Summer Internship in Data Analytics and Intelligence at the Department of Information Technology, JSS Academy of Technical Education, Noida, is an authentic record of our own work carried out under the guidance of Dr. Sukhendra Singh and Dr. Prachi Chhabra. The matter presented in this report has not been submitted by us, in part or in full, for the award of any other degree or diploma of this or any other institution.

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Date: 20 July 2026

ABSTRACT

Diabetes is one of the fastest-growing chronic diseases worldwide, and early identification of at-risk individuals remains a major public-health challenge. This project presents a **Diabetes Prediction and Risk Scoring System** built on the CDC's **BRFSS 2015 Diabetes Health Indicators Dataset**, a self-reported health survey covering more than 250,000 respondents across 21 demographic, lifestyle, and health indicators. After de-duplication and a stratified train/test split, **SMOTE** was applied to the training data only to address a severe three-class imbalance (Healthy, Prediabetic, Diabetic). Seven classifiers — **Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost, TabTransformer**, and **FT-Transformer** — were trained and evaluated using both weighted and macro-averaged metrics, with **SHAP** applied to the tree-based models for per-patient explainability. Gradient-boosted models achieved the strongest weighted performance (~ 0.81 F1), while the transformer architectures achieved the highest macro-recall and ROC-AUC, better surfacing the rare Prediabetic class. The resulting system produces a personalized 0–100 diabetes risk score alongside per-patient **SHAP** explanations, delivered through an interactive dashboard supporting both live inference and precomputed model comparison. The project's contributions include a leakage-free **SMOTE** pipeline, a systematic comparison of gradient-boosted trees against attention-based tabular transformers on this dataset, and explainable, personalized risk scoring intended to support — not replace — clinical decision-making.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Diabetes mellitus is one of the most prevalent chronic diseases affecting millions of people worldwide. It is a metabolic disorder characterized by persistently elevated blood glucose levels caused either by insufficient insulin production by the pancreas or the body's inability to effectively utilize the insulin it produces. If left undiagnosed or untreated, diabetes can lead to severe complications such as cardiovascular diseases, kidney failure, nerve damage, vision impairment, stroke, and even premature death. According to the World Health Organization (WHO) and the International Diabetes Federation (IDF), the global prevalence of diabetes has increased significantly over the past few decades due to factors such as sedentary lifestyles, unhealthy dietary habits, obesity, aging populations, and genetic predisposition. As a result, diabetes has become one of the leading public health concerns worldwide, placing a substantial burden on healthcare systems and reducing the quality of life of affected individuals.

Early detection and continuous monitoring of diabetes play a vital role in reducing disease progression and preventing long-term complications. Traditional diagnostic methods, including fasting plasma glucose tests, oral glucose tolerance tests, and glycated hemoglobin (HbA1c) tests, are clinically effective but often require laboratory testing, medical supervision, and additional healthcare resources. In many cases, individuals remain undiagnosed until noticeable symptoms or complications arise. Therefore, there is a growing need for intelligent computational systems capable of identifying individuals who are at a higher risk of developing diabetes before severe complications occur. Such systems can assist healthcare professionals in making timely clinical decisions and enable patients to adopt preventive measures at an early stage.

Recent advancements in Artificial Intelligence (AI), Machine Learning (ML), and data analytics have revolutionized the healthcare industry by enabling predictive analysis using large-scale medical datasets. Machine learning algorithms have demonstrated significant

potential in learning complex relationships between patient health indicators and disease outcomes. By analyzing historical patient records, these algorithms can identify hidden patterns, classify individuals based on disease risk, and generate accurate predictions without requiring manually defined decision rules. Consequently, machine learning has become an effective tool for developing decision-support systems capable of assisting physicians in disease diagnosis, prognosis, and treatment planning.

This project focuses on the development of a **Diabetes Prediction and Risk Scoring System** using supervised machine learning techniques. The proposed system utilizes the **Behavioral Risk Factor Surveillance System (BRFSS) 2015 Diabetes Health Indicators Dataset**, which contains over 250,000 patient records collected through health surveys. Unlike smaller clinical datasets, the BRFSS dataset provides comprehensive demographic, lifestyle, and health-related information, including body mass index (BMI), age, blood pressure, cholesterol status, physical activity, smoking habits, general health condition, mental health, physical health, income, and educational level. These attributes enable the development of a robust predictive model capable of identifying diabetes risk based on multiple health indicators rather than relying solely on laboratory test results.

In addition to predicting the diabetes category, the proposed system also estimates a **personalized diabetes risk score ranging from 0 to 100**, representing the probability of an individual developing diabetes. Rather than providing only a binary prediction, the risk score offers more meaningful information that can assist healthcare professionals and patients in understanding the severity of the predicted risk. Furthermore, the project incorporates **SHapley Additive exPlanations (SHAP)** to improve model interpretability. SHAP explains the contribution of individual health indicators toward each prediction, thereby enhancing transparency and supporting trustworthy AI-based clinical decision-making.

The primary objective of this project is not only to maximize prediction accuracy but also to develop an interpretable, reliable, and clinically meaningful machine learning framework capable of assisting healthcare practitioners in identifying high-risk individuals at an early stage. Early identification enables timely medical intervention, lifestyle modification, and preventive healthcare strategies, ultimately reducing the incidence of diabetes-related complications and improving patient outcomes.

Overall, this project demonstrates how machine learning techniques can be effectively integrated with healthcare data to build intelligent predictive systems that support disease prevention and personalized healthcare. By combining comprehensive data preprocessing, multiple classification algorithms, risk score estimation, explainable artificial intelligence, and extensive performance evaluation, the proposed system contributes toward the development of practical and data-driven solutions for modern healthcare challenges.

1.2 Background of Diabetes

Diabetes mellitus is a chronic metabolic disorder that occurs when the body is unable to produce sufficient insulin or cannot effectively utilize the insulin that it produces. Insulin is a hormone secreted by the beta cells of the pancreas, and its primary function is to regulate blood glucose levels by allowing glucose to enter body cells where it is used as a source of energy. When insulin production is inadequate or when the body's cells become resistant to insulin, glucose accumulates in the bloodstream, resulting in persistently elevated blood sugar levels, a condition known as hyperglycemia. If blood glucose levels remain uncontrolled for an extended period, diabetes can lead to severe complications affecting multiple organs of the human body.

Over the past few decades, diabetes has emerged as one of the fastest-growing non-communicable diseases worldwide. According to the International Diabetes Federation (IDF), hundreds of millions of adults currently live with diabetes, and this number continues to increase every year due to factors such as rapid urbanization, unhealthy eating habits, obesity, physical inactivity, aging populations, and genetic susceptibility. The disease not only reduces life expectancy but also imposes significant economic and healthcare burdens on individuals, families, and governments. Healthcare systems across the world spend billions of dollars annually on diabetes diagnosis, treatment, medication, hospitalization, and the management of diabetes-related complications.

Diabetes is primarily classified into three major types: **Type 1 Diabetes**, **Type 2 Diabetes**, and **Gestational Diabetes**.

Type 1 Diabetes is an autoimmune disorder in which the body's immune system mistakenly attacks and destroys the insulin-producing beta cells of the pancreas. As a result,

little or no insulin is produced, and patients require lifelong insulin therapy to survive. This type of diabetes usually develops during childhood or adolescence, although it may also occur in adults. Since the exact cause remains unknown, current treatment focuses primarily on insulin administration, continuous glucose monitoring, and lifestyle management.

Type 2 Diabetes is the most common form of diabetes, accounting for nearly 90–95% of all diabetes cases worldwide. It occurs when the body's cells become resistant to insulin or when the pancreas fails to produce sufficient insulin to meet the body's requirements. Unlike Type 1 diabetes, Type 2 diabetes develops gradually over time and is strongly associated with modifiable lifestyle factors such as obesity, lack of physical activity, unhealthy dietary patterns, smoking, excessive alcohol consumption, and chronic stress. Because its symptoms often develop slowly, many individuals remain undiagnosed until serious complications have already developed. Early detection and preventive interventions can significantly reduce disease progression and improve patient outcomes.

Gestational Diabetes Mellitus (GDM) develops during pregnancy when hormonal changes reduce the body's ability to utilize insulin effectively. Although blood glucose levels usually return to normal after childbirth, women diagnosed with gestational diabetes have a significantly higher risk of developing Type 2 diabetes later in life. Furthermore, gestational diabetes may increase the likelihood of pregnancy complications and can affect both maternal and fetal health if left untreated.

In addition to these three major categories, healthcare professionals also recognize **Prediabetes**, a condition in which blood glucose levels are higher than normal but not high enough to be classified as diabetes. Prediabetes serves as an important warning stage because individuals with this condition have a substantially increased risk of progressing to Type 2 diabetes if preventive measures are not adopted. Fortunately, lifestyle modifications such as maintaining a healthy diet, increasing physical activity, achieving weight reduction, and regular health monitoring can often delay or even prevent the progression from prediabetes to diabetes.

The Four Categories of Diabetes Recognised in This Project

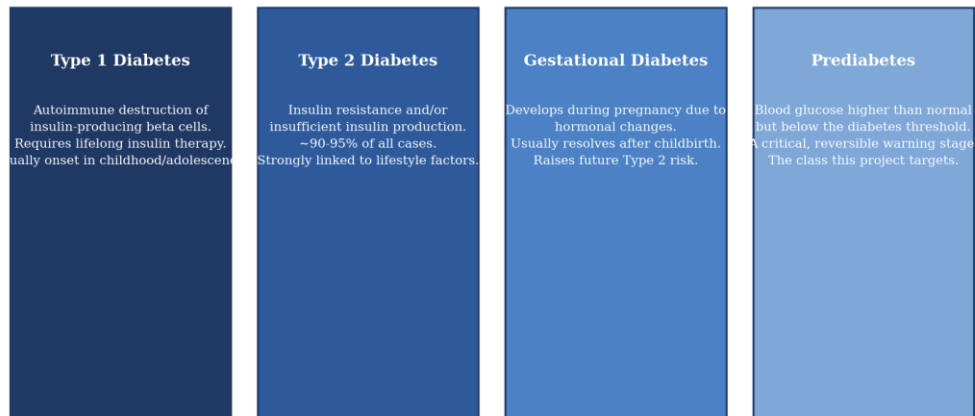


Figure 1.1 — The four categories of diabetes

The development of diabetes is influenced by multiple demographic, behavioral, genetic, and environmental factors. Some risk factors are non-modifiable, including age, family history, ethnicity, and genetic predisposition. However, many significant risk factors are modifiable and can be controlled through healthier lifestyle choices. Common modifiable risk factors include obesity, high body mass index (BMI), hypertension, elevated cholesterol levels, physical inactivity, smoking, unhealthy dietary habits, poor mental health, and excessive alcohol consumption. These factors are particularly important because they form the basis of many modern machine learning models developed for diabetes prediction.

Diabetes Risk Factors Captured by the BRFSS Feature Set

Non-Modifiable Risk Factors

Age
Family history
Ethnicity
Genetic predisposition

Modifiable Risk Factors

Obesity / High BMI
Physical inactivity
Unhealthy diet
Smoking
High blood pressure
High cholesterol
Heavy alcohol use
Poor mental health

Figure 1.2 — Modifiable and non-modifiable risk factors

The symptoms of diabetes vary depending on the severity of the disease and the individual's blood glucose levels. Common symptoms include frequent urination (polyuria), excessive thirst (polydipsia), increased hunger (polyphagia), unexplained weight loss, fatigue, blurred vision, delayed wound healing, frequent infections, numbness in the hands and feet, and reduced physical endurance. In many cases, especially Type 2 diabetes, symptoms may remain mild or unnoticed for several years, making early diagnosis particularly challenging.

If diabetes is not diagnosed and managed effectively, it can lead to serious long-term complications affecting nearly every major organ system. Persistent hyperglycemia damages blood vessels and nerves, increasing the risk of cardiovascular diseases such as heart attacks and strokes. Diabetes is also one of the leading causes of chronic kidney disease, diabetic retinopathy resulting in vision loss, peripheral neuropathy, diabetic foot ulcers, and lower-limb amputations. Additionally, uncontrolled diabetes weakens the immune system, increasing susceptibility to infections and reducing the body's ability to recover from illness.

Traditional methods for diagnosing diabetes include laboratory tests such as the **Fasting Plasma Glucose (FPG) Test**, **Oral Glucose Tolerance Test (OGTT)**, **Random Blood Sugar Test**, and **Glycated Hemoglobin (HbA1c) Test**. These diagnostic procedures are clinically reliable; however, they generally require medical supervision, laboratory facilities, and trained healthcare professionals. In many developing regions, limited healthcare accessibility and

delayed clinical screening contribute to a large number of undiagnosed diabetes cases. Consequently, researchers have increasingly focused on computational approaches capable of predicting diabetes risk using readily available demographic and lifestyle information.

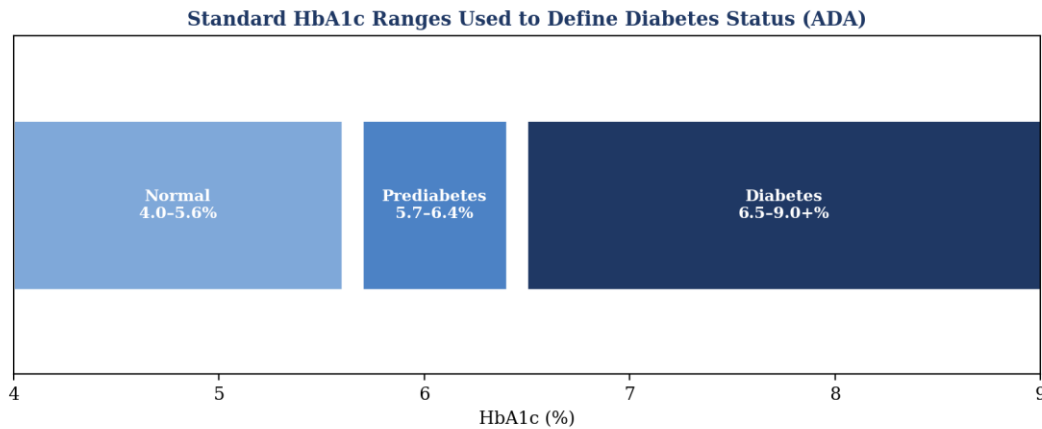


Figure 1.3 — Standard HbA1c ranges used to define diabetes status

Recent advances in Artificial Intelligence and Machine Learning have significantly transformed disease prediction by enabling computers to learn complex patterns from historical healthcare data. Instead of relying solely on laboratory measurements, machine learning algorithms can analyze multiple health indicators simultaneously, including body mass index, age, blood pressure, cholesterol status, physical activity, smoking habits, mental health, and socioeconomic factors. These algorithms can identify individuals who are more likely to develop diabetes, thereby assisting physicians in clinical decision-making and enabling preventive healthcare interventions before severe complications occur.

Considering the increasing prevalence of diabetes and the growing availability of healthcare data, intelligent predictive systems based on machine learning have become an essential component of modern healthcare. Such systems not only support physicians in early diagnosis but also empower individuals to understand their health risks, adopt healthier lifestyles, and seek timely medical consultation. Therefore, the development of accurate, interpretable, and reliable diabetes prediction models represents an important step toward improving preventive healthcare and reducing the global burden of diabetes.

1.3 Motivation

Diabetes has become one of the most significant public health challenges of the twenty-first century due to its rapidly increasing prevalence and its long-term impact on human health. Every year, millions of individuals are diagnosed with diabetes, while many others remain unaware of their condition until serious medical complications develop. The progressive nature of diabetes makes early identification extremely important because timely diagnosis and preventive interventions can significantly reduce the risk of severe complications such as cardiovascular diseases, kidney failure, blindness, nerve damage, and lower limb amputations. Despite advancements in medical technology, early diagnosis continues to remain a challenge, particularly in developing countries where access to healthcare facilities and regular medical screening is limited.

Traditional methods for diagnosing diabetes primarily rely on laboratory investigations such as fasting blood glucose tests, oral glucose tolerance tests, and HbA1c measurements. Although these diagnostic procedures are clinically accurate, they often require hospital visits, specialized laboratory equipment, trained healthcare professionals, and considerable time for testing and interpretation. As a result, many individuals delay routine health examinations, causing diabetes to remain undetected during its early stages. This delay reduces the effectiveness of treatment and increases the likelihood of irreversible health complications.

The rapid growth of healthcare data and advances in Artificial Intelligence have created new opportunities for developing intelligent disease prediction systems capable of assisting healthcare professionals in clinical decision-making. Machine learning techniques have demonstrated remarkable capabilities in extracting meaningful patterns from large-scale healthcare datasets and identifying complex relationships between multiple health indicators and disease outcomes. Unlike conventional statistical methods, machine learning algorithms can automatically learn from historical patient records and continuously improve their predictive performance as more data become available.

Another important motivation behind this project is the increasing demand for **Explainable Artificial Intelligence (XAI)** in healthcare. Although modern machine learning models often provide highly accurate predictions, many of them function as black-box systems

where the reasoning behind a prediction is not easily understandable. In clinical applications, healthcare professionals require not only accurate predictions but also clear explanations regarding the factors influencing those predictions. Therefore, integrating explainability techniques such as SHAP (SHapley Additive exPlanations) enhances transparency, builds trust in machine learning systems, and supports informed clinical decision-making.

This project is also motivated by the need to develop a practical decision-support system that not only predicts whether an individual is diabetic but also estimates a personalized diabetes risk score. Such a system enables individuals to understand their level of risk and encourages timely lifestyle modifications, regular medical consultations, and preventive healthcare practices. Rather than replacing medical professionals, the proposed system is intended to function as an intelligent assistant capable of supporting physicians by providing rapid, data-driven predictions based on patient health indicators.

Furthermore, comparing multiple supervised machine learning algorithms provides valuable insights into their strengths, limitations, and suitability for diabetes prediction. Since different algorithms learn patterns in different ways, evaluating multiple models allows the identification of the most reliable classifier for the given dataset while improving the overall robustness of the proposed framework.

Therefore, the primary motivation of this project is to utilize machine learning techniques for developing an accurate, interpretable, and efficient diabetes prediction system capable of assisting healthcare professionals, supporting preventive medicine, reducing diagnostic delays, and ultimately improving patient outcomes through early risk assessment.

1.4 Problem Statement

Diabetes has become a major global healthcare concern due to its continuously increasing incidence and the serious medical complications associated with delayed diagnosis. Millions of individuals remain undiagnosed because the disease often develops gradually and may not exhibit noticeable symptoms during its early stages. Consequently, many patients seek medical attention only after experiencing severe complications such as cardiovascular disorders, kidney damage, diabetic retinopathy, neuropathy, or other chronic conditions. Early identification of

individuals at high risk of developing diabetes is therefore essential for implementing preventive healthcare measures and reducing disease progression.

Conventional diabetes diagnosis primarily depends on laboratory investigations, including fasting plasma glucose tests, oral glucose tolerance tests, and glycated hemoglobin (HbA1c) measurements. While these diagnostic methods are clinically reliable, they require specialized medical infrastructure, laboratory facilities, trained healthcare personnel, and regular clinical visits. In many healthcare settings, especially in rural and resource-limited regions, access to these facilities may be limited, leading to delayed diagnosis and reduced opportunities for early intervention.

At the same time, healthcare organizations continuously collect large volumes of patient data containing valuable information related to demographic characteristics, lifestyle habits, medical history, and behavioral risk factors. However, much of this data remains underutilized due to the absence of efficient predictive systems capable of extracting meaningful knowledge from complex healthcare datasets. Traditional statistical techniques often struggle to model the nonlinear relationships that exist among multiple health indicators influencing diabetes risk.

Although numerous machine learning models for diabetes prediction have been proposed in recent years, many existing approaches suffer from several limitations. A significant number of studies have been conducted using relatively small datasets, such as the Pima Indians Diabetes Dataset, which contains only a limited number of patient records. Models trained on such datasets may exhibit reduced generalization capability when applied to larger and more diverse populations. Furthermore, several existing studies focus primarily on maximizing classification accuracy while providing limited emphasis on model interpretability, comprehensive feature analysis, or personalized risk estimation.

Another challenge is that healthcare professionals require predictive systems that are not only accurate but also transparent and explainable. Black-box machine learning models may achieve high prediction accuracy, yet their inability to explain individual predictions reduces trust and limits their practical adoption in clinical environments. Consequently, integrating explainable artificial intelligence techniques into predictive healthcare systems has become increasingly important.

The successful implementation of this project is expected to assist healthcare professionals in identifying high-risk individuals at an early stage, facilitate timely preventive interventions, support evidence-based medical decision-making, and contribute toward reducing the long-term burden of diabetes through intelligent healthcare analytics.

1.5 Objectives of the Project

The primary objective of this project is to develop an intelligent and reliable machine learning-based system capable of accurately predicting diabetes using demographic, behavioral, and health-related indicators. The proposed system aims to assist healthcare professionals in identifying individuals at risk of developing diabetes at an early stage, thereby enabling timely medical intervention and reducing the likelihood of severe complications.

The specific objectives of this project are as follows:

To study the causes, risk factors, symptoms, and global impact of diabetes mellitus and understand the importance of early disease prediction.

To collect and analyze the **Behavioral Risk Factor Surveillance System (BRFSS) 2015 Diabetes Health Indicators Dataset** containing demographic, lifestyle, and medical information of individuals.

To perform comprehensive data preprocessing by handling missing values, checking duplicate records, validating data quality, and preparing the dataset for machine learning model development.

To conduct Exploratory Data Analysis (EDA) using statistical summaries and visualization techniques such as histograms, count plots, box plots, correlation heatmaps, and pair plots in order to understand feature distributions and identify significant relationships among variables.

To implement and compare multiple supervised machine learning classification algorithms, including Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, Support Vector Machine (SVM), and Gaussian Naïve Bayes, for diabetes prediction.

To evaluate the performance of each classification model using standard performance metrics such as Accuracy, Precision, Recall, F1-Score, Confusion Matrix, Receiver Operating Characteristic (ROC) Curve, and Area Under the Curve (AUC).

To identify the most effective machine learning algorithm by comparing the predictive performance of all implemented models under identical experimental conditions.

To estimate a personalized diabetes risk score based on the predicted probability generated by the selected machine learning model, enabling better interpretation of an individual's likelihood of developing diabetes.

To improve model transparency and interpretability by applying Explainable Artificial Intelligence (XAI) techniques such as SHapley Additive exPlanations (SHAP), thereby identifying the contribution of each health indicator toward the final prediction.

To develop a reliable, interpretable, and data-driven decision-support system that can assist healthcare professionals in early diagnosis, preventive healthcare planning, and evidence-based clinical decision-making.

1.6 Scope of the Project

The scope of this project is focused on designing and implementing a machine learning-based diabetes prediction system using publicly available healthcare data. The proposed system utilizes the **Behavioral Risk Factor Surveillance System (BRFSS) 2015 Diabetes Health Indicators Dataset**, which contains a large collection of demographic, lifestyle, and health-related information collected through national health surveys. The dataset provides valuable indicators such as body mass index (BMI), age, blood pressure, cholesterol status, smoking habits, physical activity, mental health, physical health, education level, income, and general health status, enabling comprehensive analysis for diabetes prediction.

The project primarily focuses on the application of supervised machine learning algorithms for disease prediction rather than replacing traditional clinical diagnosis. Multiple classification algorithms are implemented and evaluated to determine the most suitable model for predicting diabetes based on patient health indicators. Extensive data preprocessing and

exploratory data analysis are performed to improve data quality and understand relationships among various features before model development.

Furthermore, the proposed framework generates a personalized diabetes risk score that provides a quantitative estimate of an individual's probability of developing diabetes. Such information can assist healthcare professionals in identifying high-risk patients and recommending appropriate preventive measures.

Although the proposed system demonstrates the practical application of machine learning in healthcare, it is intended to function only as a clinical decision-support tool and not as a replacement for medical diagnosis performed by qualified healthcare professionals. The prediction results should always be interpreted alongside clinical examinations and laboratory investigations.

The scope of this project is limited to the BRFSS healthcare dataset and supervised machine learning techniques. Future work may include integrating deep learning models, real-time patient monitoring through wearable devices, cloud-based healthcare platforms, electronic health record (EHR) integration, and mobile health applications to further enhance prediction accuracy and accessibility.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

Diabetes mellitus has become one of the most prevalent chronic diseases worldwide and continues to pose significant challenges to global healthcare systems. The increasing number of diabetes cases, coupled with the serious complications associated with delayed diagnosis, has motivated researchers to develop intelligent prediction systems capable of identifying individuals at risk during the early stages of the disease. Early prediction enables timely medical intervention, lifestyle modification, and preventive healthcare strategies, thereby reducing the incidence of severe complications such as cardiovascular diseases, kidney failure, diabetic neuropathy, retinopathy, and stroke.

In recent years, rapid advancements in Artificial Intelligence (AI), Machine Learning (ML), and data analytics have significantly transformed the field of healthcare. Machine learning algorithms have demonstrated exceptional capability in analyzing large-scale healthcare datasets and discovering complex relationships among various demographic, physiological, behavioral, and lifestyle factors that contribute to disease development. Unlike traditional statistical methods, machine learning models can automatically learn hidden patterns from historical patient records and generate accurate predictions without explicitly programmed decision rules.

A considerable amount of research has been conducted to develop diabetes prediction models using various machine learning algorithms, including Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, Naïve Bayes, Artificial Neural Networks, K-Nearest Neighbors, Gradient Boosting, and ensemble learning techniques. These studies have employed different datasets, preprocessing methods, feature engineering techniques, and evaluation metrics to improve prediction accuracy and model robustness. Many researchers have also investigated data preprocessing strategies such as missing value imputation, outlier detection, feature scaling, feature selection, and dimensionality reduction to enhance classifier performance.

Although significant progress has been achieved in diabetes prediction, several research challenges still remain. Many existing studies rely on relatively small datasets, limiting the generalization capability of the developed models. Some research primarily focuses on maximizing classification accuracy while paying limited attention to model interpretability, feature importance analysis, or personalized risk assessment. Furthermore, black-box prediction models often fail to provide sufficient explanation regarding the factors responsible for a particular prediction, thereby reducing trust among healthcare professionals.

Therefore, a comprehensive review of previously published research is essential to understand the strengths and limitations of existing methodologies, identify current research gaps, and justify the development of the proposed diabetes prediction framework. The literature review presented in this chapter examines recent studies related to diabetes prediction using machine learning techniques. It compares different datasets, preprocessing approaches, classification algorithms, performance evaluation methods, and research outcomes. The insights obtained from these studies provide the foundation for designing the methodology adopted in the proposed project.

The objective of this literature review is not only to summarize previous research but also to critically analyze existing approaches in order to identify opportunities for further improvement. The findings of this chapter serve as the basis for selecting suitable machine learning algorithms, preprocessing techniques, evaluation metrics, and explainability methods implemented in the proposed diabetes prediction and risk scoring system.

2.2 Review of Existing Literature

Machine learning has become one of the most widely adopted approaches for disease prediction due to its ability to analyze complex healthcare data and identify hidden relationships among multiple clinical and demographic variables. Numerous researchers have proposed different machine learning frameworks for diabetes prediction using various public healthcare datasets. These studies differ in terms of dataset size, preprocessing techniques, feature engineering methods, classification algorithms, evaluation metrics, and prediction performance.

Most of the existing research focuses on the **Pima Indians Diabetes Dataset (PID)**, which has become one of the most commonly used benchmark datasets for diabetes prediction. The dataset contains medical information of female patients, including glucose concentration, blood pressure, insulin level, body mass index, diabetes pedigree function, age, and pregnancy count. Researchers have extensively applied supervised machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Naïve Bayes, Artificial Neural Networks (ANN), and ensemble learning techniques on this dataset to improve prediction accuracy.

Several studies have demonstrated that appropriate data preprocessing significantly improves prediction performance. Common preprocessing techniques include handling missing values, removing outliers, feature scaling, normalization, feature selection, and cross-validation. Proper preprocessing helps machine learning algorithms learn meaningful patterns while reducing the impact of noisy or inconsistent data. Researchers have also compared different feature selection techniques such as correlation analysis, Principal Component Analysis (PCA), Independent Component Analysis (ICA), and recursive feature elimination to identify the most influential health indicators associated with diabetes.

In addition to traditional machine learning algorithms, recent research has increasingly explored ensemble learning methods that combine multiple classifiers to improve predictive performance. Ensemble techniques such as Random Forest, AdaBoost, Gradient Boosting, XGBoost, and voting classifiers have consistently demonstrated higher prediction accuracy and robustness than individual machine learning models. By aggregating predictions from multiple classifiers, ensemble models reduce variance, improve generalization capability, and achieve better performance on unseen patient data.

Recent advancements have also introduced deep learning approaches, particularly Artificial Neural Networks (ANNs) and Multilayer Perceptrons (MLPs), for diabetes prediction. These models are capable of learning complex nonlinear relationships among healthcare variables and often achieve competitive performance when trained on sufficiently large datasets. However, deep learning models generally require larger computational resources and larger datasets compared to conventional machine learning algorithms.

Another emerging trend in diabetes prediction research is the integration of Explainable Artificial Intelligence (XAI). Although many advanced machine learning models achieve excellent prediction accuracy, they often function as black-box systems that provide little explanation regarding individual predictions. To address this challenge, researchers have started incorporating explainability techniques such as SHAP (SHapley Additive exPlanations), LIME (Local Interpretable Model-Agnostic Explanations), and feature importance analysis. These methods improve transparency by identifying the contribution of individual health indicators toward the final prediction, thereby increasing trust among healthcare professionals.

Despite the significant progress achieved in recent years, several limitations continue to exist in the current literature. Many published studies rely on relatively small datasets that may not adequately represent diverse patient populations. Some studies focus primarily on improving prediction accuracy while providing limited analysis of model interpretability, personalized risk estimation, or real-world clinical applicability. Furthermore, only a limited number of studies have utilized large-scale healthcare survey datasets that include demographic, socioeconomic, lifestyle, and behavioral factors together with clinical indicators.

To address these limitations, the present project utilizes the **Behavioral Risk Factor Surveillance System (BRFSS) 2015 Diabetes Health Indicators Dataset**, which contains more than **250,000 patient records**. The project performs comprehensive data preprocessing, exploratory data analysis, comparative evaluation of multiple supervised machine learning algorithms, personalized diabetes risk score estimation, and SHAP-based explainability. By combining robust preprocessing, multiple classification models, extensive performance evaluation, and explainable artificial intelligence, the proposed framework aims to provide a more reliable, interpretable, and practical diabetes prediction system suitable for healthcare decision support.

2.3 Review of Research Papers

2.3.1 Paper 1

Diabetes Prediction Using Machine Learning Algorithms and Ontology (2022)

Authors: Hakim El Massari, Zineb Sabouri, Sajida Mhammedi, and Noredine Gherabi

Available at: <https://www.semanticscholar.org/paper/Diabetes-Prediction-Using-Machine-Learning-and-Massari-Sabouri/00f13c1bfe71905e9f1133e4b3f06861f18e0b14>

This paper presents a comparative study between conventional machine learning algorithms and ontology-based machine learning techniques for diabetes prediction. The objective of the study was to determine whether ontology can improve prediction accuracy while providing better interpretability than traditional machine learning classifiers. The authors used the **Pima Indians Diabetes Dataset (PIDD)** and implemented several classification algorithms including Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Logistic Regression, Artificial Neural Network (ANN), Decision Tree, and Naïve Bayes. The performance of each classifier was evaluated using Accuracy, Precision, Recall, and F-Measure.

The experimental results showed that the ontology-based classifier achieved the highest accuracy of approximately **77.5%**, followed by Support Vector Machine and Logistic Regression. The authors concluded that ontology improves knowledge representation and makes predictions easier to interpret, which is particularly beneficial in healthcare applications. However, the study was conducted only on the relatively small Pima Indians Diabetes Dataset, limiting its applicability to larger populations. The concepts of explainability presented in this paper influenced the proposed project, where SHAP explainability is incorporated to improve model transparency.

2.3.2 Paper 2

Diabetes Prediction Using Ensembling of Different Machine Learning Classifiers (2020)

Authors: Md. Kamrul Hasan, Md. Ashraful Alam, Dola Das, Eklas Hossain, and Mahmudul Hasan

Available at:

https://www.researchgate.net/publication/340879584_Diabetes_Prediction_Using_Ensembling_of_Different_Machine_Learning_Classifiers

This research proposed a robust ensemble learning framework for diabetes prediction by combining multiple machine learning classifiers. Before model training, the authors performed extensive preprocessing including missing value handling, outlier removal, feature standardization, feature selection, and K-fold cross-validation. The implemented classifiers included KNN, Decision Tree, Random Forest, AdaBoost, Naïve Bayes, XGBoost, and Multilayer Perceptron (MLP). A weighted ensemble model was then developed using AUC-based weights to improve overall prediction performance.

The proposed ensemble classifier achieved an **AUC of 0.950**, outperforming individual classifiers and improving prediction accuracy compared with previous studies. The major contribution of this work is the use of weighted ensemble learning and systematic preprocessing techniques. However, the framework was developed using only the Pima Indians Diabetes Dataset and did not include explainability techniques or personalized risk estimation. The methodology adopted in this project is inspired by this research through the comparison of multiple machine learning algorithms and comprehensive model evaluation.

2.3.3 Paper 3

Supervised Machine Learning Based Ensemble Model for Accurate Prediction of Type 2 Diabetes (2019)

Authors: Ramya Akula, Ni Nguyen, and Ivan Garibay

Available at: <https://ieeexplore.ieee.org/document/9020358>

The authors proposed a supervised ensemble learning model for predicting Type 2 diabetes using electronic health records obtained from the Practice Fusion database. The dataset contained nearly **10,000 patient records** including demographic information, blood pressure, height, weight, and other clinical measurements. Several machine learning algorithms such as K-Nearest Neighbors, Support Vector Machine, Decision Tree, Random Forest, Gradient Boosting, Neural Network, and Naïve Bayes were evaluated individually before developing a weighted ensemble model.

The proposed ensemble model achieved approximately **85% prediction accuracy**, outperforming most individual classifiers. The authors concluded that combining multiple

classifiers improves prediction reliability and reduces misclassification. However, the study mainly focused on improving accuracy and provided limited explanation regarding feature importance and model interpretability. This paper demonstrates the effectiveness of ensemble learning in healthcare prediction systems and supports the idea of comparing multiple classification algorithms before selecting the best-performing model.

2.3.4 Paper 4

Predicting Diabetes Using Machine Learning: A Comparative Study of Classifiers (2025)

Authors: Mahade Hasan and Farhana Yasmin

Available at: <https://arxiv.org/abs/2505.07036>

This paper presents a comparative analysis of various machine learning algorithms used for diabetes prediction. The authors reviewed several existing prediction models developed using algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, Naïve Bayes, Artificial Neural Networks, and ensemble learning techniques. The study also discussed the importance of feature engineering, preprocessing, dimensionality reduction, and performance evaluation using metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

The paper concluded that Random Forest, ensemble models, and Support Vector Machine consistently produce competitive performance for diabetes prediction. The authors also emphasized the importance of using larger and more diverse datasets and incorporating explainable artificial intelligence to improve clinical adoption. The findings support the methodology adopted in the proposed project, which utilizes a large healthcare dataset, multiple classifiers, SHAP explainability, and personalized risk score estimation.

2.3.5 Paper 5

Prediction of Type 2 Diabetes Using Machine Learning Classification Methods (2020)

Authors: N. P. Tigga and S. Garg

This study investigated the application of different supervised machine learning algorithms for the prediction of Type 2 diabetes. The authors implemented classification models including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbors, and Naïve Bayes on healthcare data and compared their predictive performance. The research highlighted that appropriate preprocessing, feature selection, and parameter optimization significantly improve classification accuracy.

The study reported that ensemble-based approaches and Random Forest classifiers generally achieved better predictive performance than traditional classification methods. The authors concluded that machine learning techniques can assist healthcare professionals in early diabetes diagnosis and reduce the burden of manual screening. However, they also recommended using larger datasets and more interpretable prediction models for practical healthcare applications. This study further supports the selection of multiple machine learning algorithms in the proposed project and reinforces the importance of comparative model evaluation for identifying the most reliable classifier.

2.4 Comparative Analysis of Existing Studies

A comprehensive comparison of the reviewed research papers reveals that machine learning has become an effective approach for diabetes prediction, with different studies employing various datasets, preprocessing techniques, classification algorithms, and evaluation metrics. Although the primary objective of these studies is to improve prediction accuracy, each research work follows a different methodology and exhibits unique strengths and limitations.

Most of the reviewed studies utilized the **Pima Indians Diabetes Dataset (PIDD)** because of its simplicity and widespread availability. While this dataset has been extensively used as a benchmark for evaluating diabetes prediction algorithms, its relatively small size limits the generalization capability of the developed models when applied to larger and more diverse populations. In contrast, the proposed project utilizes the **Behavioral Risk Factor Surveillance System (BRFSS) 2015 Diabetes Health Indicators Dataset**, which contains more than **250,000 patient records**, making the prediction model more robust and suitable for real-world healthcare applications.

The literature also indicates that proper data preprocessing plays a significant role in improving prediction performance. Almost every reviewed study performed preprocessing tasks such as missing value handling, feature scaling, normalization, outlier removal, or feature selection before model training. These preprocessing techniques help improve data quality and enable machine learning algorithms to learn meaningful relationships among patient health indicators. Similarly, the proposed project performs comprehensive preprocessing, exploratory data analysis, and feature preparation before developing the prediction models.

Another important observation from the reviewed literature is the extensive use of supervised machine learning algorithms. Logistic Regression, Decision Tree, Support Vector Machine (SVM), Random Forest, K-Nearest Neighbors (KNN), Naïve Bayes, Artificial Neural Networks (ANN), and ensemble learning techniques were the most commonly implemented classifiers. Among these methods, ensemble learning approaches generally achieved better prediction performance because they combine the strengths of multiple classifiers while reducing the weaknesses of individual models.

Recent studies have also emphasized the importance of model interpretability in healthcare applications. Although several machine learning models achieve high prediction accuracy, many of them behave as black-box systems and provide limited explanation regarding the factors influencing individual predictions. To overcome this limitation, researchers have started incorporating Explainable Artificial Intelligence (XAI) techniques such as SHAP and LIME to improve transparency and increase trust among healthcare professionals.

Based on the reviewed literature, it can be concluded that an ideal diabetes prediction system should combine comprehensive data preprocessing, multiple machine learning algorithms, large-scale healthcare datasets, extensive performance evaluation, and explainable artificial intelligence techniques. The proposed project follows this integrated approach by utilizing the BRFSS healthcare dataset, implementing multiple supervised learning algorithms, generating personalized diabetes risk scores, and incorporating SHAP-based explainability to support accurate and interpretable healthcare decision-making.

Table 2.1 Comparative Analysis of Reviewed Research Papers

Paper	Dataset Used	Algorithms	Major Contribution	Limitations
Diabetes Prediction Using Machine Learning Algorithms and Ontology (2022)	Pima Indians Diabetes Dataset	SVM, KNN, Logistic Regression, ANN, Naïve Bayes, Ontology	Improved prediction interpretability using ontology	Small dataset, moderate accuracy
Diabetes Prediction Using Ensembling of Different Machine Learning Classifiers (2020)	Pima Indians Diabetes Dataset	KNN, Random Forest, AdaBoost, XGBoost, MLP, Ensemble	Weighted ensemble achieved high prediction accuracy	No explainability, small dataset
Supervised Machine Learning Based Ensemble Model for Accurate Prediction of Type 2 Diabetes (2019)	Electronic Health Records	Random Forest, SVM, Gradient Boosting, ANN, Ensemble	Ensemble learning improved prediction reliability	Limited model interpretation
Predicting Diabetes Using Machine Learning: A Comparative Study of Classifiers (2025)	Diabetes Healthcare Dataset	Logistic Regression, SVM, Decision Tree, Random Forest	Comparative evaluation of multiple classifiers	Limited personalized risk assessment
Prediction of Type 2 Diabetes Using Machine Learning Classification Methods (2020)	Healthcare Dataset	Logistic Regression, Random Forest, KNN, SVM, Naïve Bayes	Performance comparison of classification algorithms	Lack of explainability and large-scale validation

2.5 Research Gap

The literature survey indicates that significant progress has been made in the field of diabetes prediction using machine learning techniques. Researchers have successfully applied various supervised learning algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Naïve Bayes, Artificial Neural Networks (ANN), and ensemble learning methods for predicting diabetes. Many studies have reported satisfactory prediction accuracy and demonstrated the potential of machine learning in supporting early disease diagnosis. However, despite these advancements, several research gaps continue to exist.

One of the major limitations observed in the reviewed literature is the extensive dependence on relatively small datasets, particularly the **Pima Indians Diabetes Dataset (PIDD)**. Although this dataset is widely accepted as a benchmark dataset for diabetes prediction, it contains only **768 patient records**, which limits the generalization capability of the developed models. Models trained on such limited datasets may not accurately represent diverse populations with varying demographic, behavioral, and socioeconomic characteristics.

Another important research gap is the limited utilization of comprehensive healthcare survey datasets. Most existing studies focus primarily on clinical measurements such as glucose concentration, insulin level, blood pressure, and body mass index, while comparatively fewer studies incorporate lifestyle, behavioral, demographic, and socioeconomic factors that significantly influence diabetes risk. As a result, many prediction models fail to capture the complete health profile of individuals.

The literature review also reveals that many researchers concentrate primarily on maximizing prediction accuracy while providing limited emphasis on model interpretability. Advanced machine learning algorithms often behave as black-box systems, making it difficult for healthcare professionals to understand the reasoning behind individual predictions. This lack of transparency reduces trust and limits the practical adoption of artificial intelligence in healthcare environments where explainability is essential for clinical decision-making.

Furthermore, most existing studies provide only a categorical prediction indicating whether a patient is diabetic or non-diabetic. Very few studies attempt to generate a

personalized diabetes risk score that quantifies the probability of developing diabetes. Such risk estimation can provide more meaningful information to both healthcare professionals and patients by supporting preventive healthcare planning and early lifestyle interventions.

Another limitation observed in previous research is the absence of comprehensive comparative analysis across multiple machine learning algorithms under identical experimental conditions. Several studies evaluate only one or two classification models, making it difficult to identify the most suitable algorithm for diabetes prediction. Additionally, limited attention has been given to standardized performance evaluation using multiple metrics such as Accuracy, Precision, Recall, F1-Score, ROC Curve, and Area Under the Curve (AUC).

To address these research gaps, the proposed project utilizes the **Behavioral Risk Factor Surveillance System (BRFSS) 2015 Diabetes Health Indicators Dataset**, which contains more than **250,000 patient records** representing a large and diverse population. The project performs comprehensive data preprocessing, exploratory data analysis, and comparative evaluation of multiple supervised machine learning algorithms. In addition to predicting diabetes, the proposed framework generates a personalized diabetes risk score and incorporates **SHAP** to improve model transparency and interpretability. These enhancements aim to develop a reliable, accurate, and explainable diabetes prediction system that can effectively support healthcare professionals in early diagnosis and preventive decision-making.

Research Gap Analysis:

Limitations in Existing Research	Proposed Solution in This Project
Most studies use the small Pima Indians Diabetes Dataset (768 records).	Uses the BRFSS 2015 dataset containing over 250,000 patient records for better generalization.
Focus mainly on clinical parameters with limited lifestyle and demographic information.	Incorporates demographic, behavioral, lifestyle, and health-related features for comprehensive prediction.
Many models function as black-box systems with limited explainability.	Integrates SHAP Explainable AI to interpret feature contributions and improve transparency.

Limitations in Existing Research	Proposed Solution in This Project
Existing systems generally provide only diabetes classification.	Generates a personalized diabetes risk score (0–100) along with prediction.
Limited comparison of multiple machine learning algorithms.	Implements and compares Logistic Regression, KNN, Decision Tree, Random Forest, SVM, and Gaussian Naïve Bayes using standardized evaluation metrics.
Limited performance evaluation using comprehensive metrics.	Evaluates models using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, ROC Curve, and AUC.

2.6 Chapter Summary

This chapter presented a comprehensive review of the existing literature related to diabetes prediction using machine learning techniques. Various research studies published in recent years were analyzed to understand the methodologies, datasets, preprocessing techniques, machine learning algorithms, and evaluation metrics adopted by different researchers. The literature review highlighted the increasing importance of Artificial Intelligence and Machine Learning in the healthcare domain, particularly for the early prediction and diagnosis of diabetes.

The reviewed studies demonstrated that supervised machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Naïve Bayes, Artificial Neural Networks (ANN), and ensemble learning techniques have been successfully applied for diabetes prediction. Many researchers achieved satisfactory prediction accuracy by employing appropriate data preprocessing techniques, feature selection methods, and model optimization strategies. The literature also indicated that ensemble learning approaches generally outperform individual classifiers due to their ability to combine the strengths of multiple machine learning models.

The review further revealed that although considerable progress has been made in diabetes prediction, several limitations still exist. Most existing studies utilize relatively small datasets such as the Pima Indians Diabetes Dataset, which restricts the generalization capability of the developed models. Additionally, many prediction systems primarily focus on improving

classification accuracy while providing limited attention to model interpretability, personalized risk estimation, and the use of large-scale healthcare datasets. The lack of explainability in several machine learning models also limits their practical adoption in real-world clinical environments.

Based on these observations, the research gaps identified in this chapter provide the motivation for the proposed work. The present project addresses these limitations by utilizing the **Behavioral Risk Factor Surveillance System (BRFSS) 2015 Diabetes Health Indicators Dataset**, which contains more than **250,000 patient records**. The proposed framework performs comprehensive data preprocessing, exploratory data analysis, comparative evaluation of multiple supervised machine learning algorithms, personalized diabetes risk score estimation, and SHAP-based Explainable Artificial Intelligence (XAI) to improve prediction transparency and reliability.

Overall, the literature survey establishes a strong foundation for the proposed research and justifies the selection of machine learning techniques adopted in this project. The findings of this chapter serve as the basis for designing the methodology presented in the next chapter, where the complete framework for diabetes prediction and risk assessment is discussed in detail.

CHAPTER 3

DATASET & METHODOLOGY

3.1 Source and Overview

The data used in this project comes from the CDC's Behavioral Risk Factor Surveillance System (BRFSS), an annual telephone survey that has been run continuously since 1984 and collects health-behaviour and chronic-condition data from several hundred thousand adults across the United States every year. The specific file used is the cleaned 2015 extract distributed on Kaggle (originally compiled by Alex Teboul from the CDC's raw BRFSS microdata and mirrored on the UCI Machine Learning Repository), which reduces the original survey's 330-odd questions down to 21 health indicator features plus a three-class target variable, covering 253,680 respondents.

The target column, `Diabetes_012`, encodes three classes: 0 for no diabetes (including gestational-only diabetes during pregnancy), 1 for prediabetes, and 2 for diabetes. This three-class formulation was kept deliberately rather than collapsing it into the more commonly used binary version of the dataset, because the binary framing throws away exactly the group — prediabetics — that a screening tool is most useful for catching early.

3.2 Duplicate Removal

A known property of this dataset, noted in several of the papers reviewed above, is that a large number of rows are exact duplicates of each other. This is a structural artefact of the survey rather than a data-entry error: because every feature is a bucketed categorical or low-cardinality response (age brackets, five-point health-rating scales, yes/no lifestyle questions), it is statistically inevitable that many unrelated respondents will answer every single question identically. Left in place, these duplicates can end up split across the train and test sets, letting the model effectively "see" a test example during training in a different disguise, which quietly inflates evaluation scores.

For this project, exact duplicate rows were identified and removed prior to any modelling. This dropped roughly 24,000 rows from the original 253,680, leaving 229,781

usable, de-duplicated records. All subsequent splitting, resampling, tuning and evaluation described in this report operate on this de-duplicated set.

3.3 Class Distribution and the Imbalance Problem

The class distribution after de-duplication is heavily skewed:

Class	Label	Approx. share of data
0	Healthy (no diabetes)	$\approx 84\%$
2	Diabetic	$\approx 14\%$
1	Prediabetic	$\approx 1.8\%$

Class Distribution After De-duplication (n = 229,781)

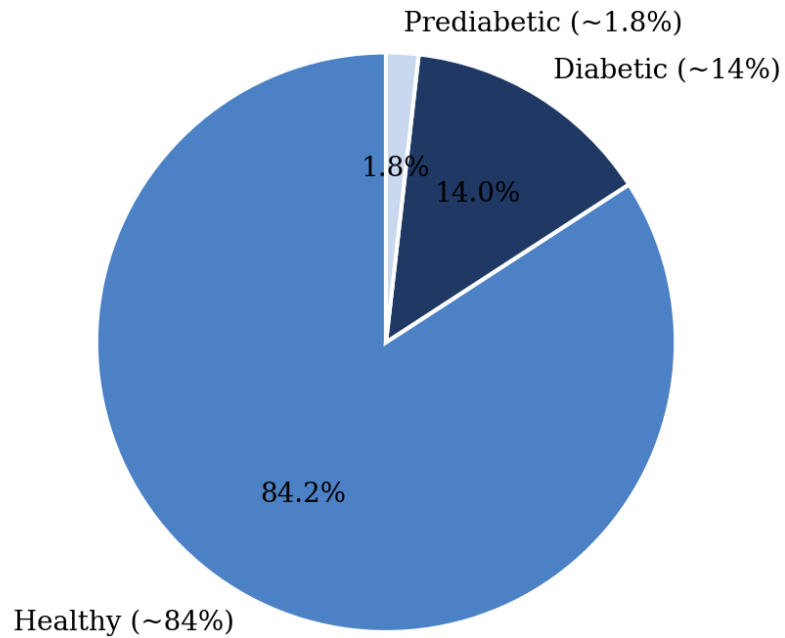


Figure 3.1 — Class distribution after de-duplication

This imbalance is the single most consequential fact about the dataset and it drives most of the design decisions described in Section 4 — the choice to apply SMOTE, the choice to score hyperparameter search on F1 rather than accuracy, and the choice to report macro-averaged metrics alongside weighted ones rather than leaning on overall accuracy as the headline number. A model that simply predicted "Healthy" for every single respondent would already be right about 84% of the time and look deceptively competent on a plain accuracy scoreboard, while being completely useless for the prediabetic group the project actually cares about.

3.4 Feature Description

The 21 predictor features fall into four broad groups: clinical/self-reported health conditions, lifestyle and behavioural factors, general health perception, and demographic/socioeconomic indicators. None of them require a blood draw or clinical measurement beyond what a person could plausibly self-report or a home device (a scale, a blood-pressure cuff) could provide.

Feature	Type	Description
HighBP	Binary	Ever told they have high blood pressure
HighChol	Binary	Ever told they have high cholesterol
CholCheck	Binary	Cholesterol check within the last 5 years
BMI	Continuous	Body Mass Index
Smoker	Binary	Smoked at least 100 cigarettes in lifetime
Stroke	Binary	Ever told they had a stroke
HeartDiseaseorAttack	Binary	Coronary heart disease or myocardial infarction history
PhysActivity	Binary	Physical activity in past 30 days (non-job related)
Fruits	Binary	Consumes fruit one or more times per day
Veggies	Binary	Consumes vegetables one or more times per day
HvyAlcoholConsump	Binary	Heavy alcohol consumption (adult thresholds)
AnyHealthcare	Binary	Has any kind of healthcare coverage

Feature	Type	Description
NoDocbcCost	Binary	Needed a doctor in past year but cost was a barrier
GenHlth	Ordinal (1–5)	Self-rated general health, excellent to poor
MentHlth	Continuous (days)	Days of poor mental health in past 30 days
PhysHlth	Continuous (days)	Days of poor physical health in past 30 days
DiffWalk	Binary	Serious difficulty walking or climbing stairs
Sex	Binary	Male / Female
Age	Ordinal (13 brackets)	Age category, 18–24 up to 80+
Education	Ordinal (1–6)	Highest level of education completed
Income	Ordinal (1–8)	Household income bracket

Two features — BMI, MentHlth and PhysHlth — are effectively continuous (the latter two are day-counts bounded at 30) and were treated as numeric inputs throughout; everything else is binary or ordinal-coded and was left in its native integer encoding for the tree-based models, which handle ordinal splits natively, and passed through the feature-tokenizer / embedding layers described in Section 5 for the two transformer models.

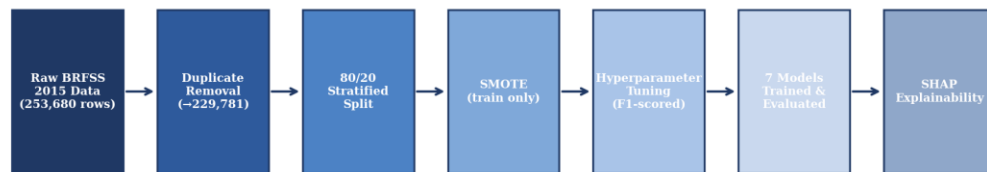


Figure 3.2 — End-to-end methodology pipeline

The methodology follows a fairly standard supervised-learning pipeline, but with three deliberate choices that depart from a lot of the published work on this dataset: the train/test split

happens before any resampling, hyperparameter tuning is scored on F1 rather than accuracy, and evaluation is reported on both macro and weighted metrics side by side rather than picking whichever one looks better.

3.5 Preprocessing

The de-duplicated dataset (229,781 rows) required very little cleaning beyond the duplicate removal itself, since the Kaggle/CDC release is already a curated extract with no missing values and consistent categorical coding. Preprocessing was limited to: casting BMI, MentHlth and PhysHlth to numeric types; verifying that all categorical/ordinal codes fell within their documented ranges (e.g. Age in 1–13, GenHlth in 1–5) with no stray values; and standard-scaling the three continuous features for the logistic regression baseline and the two transformer models, which are sensitive to feature scale in a way tree ensembles are not. Tree-based models (Random Forest, XGBoost, LightGBM, CatBoost) were trained on the unscaled features, since splitting on a threshold is invariant to monotonic rescaling.

3.6 Train/Test Split

An 80/20 stratified split was taken first, before any oversampling, preserving the roughly 84/1.8/14 class ratio in both partitions. Stratification matters here specifically because of the prediabetic class — with only about 1.8% of rows belonging to it, a non-stratified split could easily leave the test set with an even more skewed (or, by chance, more favourable) sample than the training set, making the reported metrics unstable and hard to compare across models run at different times.

3.7 Handling Class Imbalance: SMOTE, Applied Correctly

SMOTE (Synthetic Minority Over-sampling Technique) generates synthetic minority-class examples by interpolating between a real minority sample and its nearest neighbours in feature space, rather than simply duplicating existing rows (which is what plain random oversampling does, and which tends to make models overfit to the exact duplicated points). SMOTE was applied to bring the Prediabetic and Diabetic classes up to a more workable ratio relative to the Healthy class within the training set only.

The ordering here is the important part: the 80/20 split happens first, and SMOTE is fit and applied only on the resulting training partition. The test set is left in its original, untouched, real-world-imbalanced state throughout. This avoids a leakage pattern visible in several of the papers surveyed in Section 2, where oversampling applied to the whole dataset before splitting lets synthetic points derived from a test-set patient end up in the training fold — the model then gets evaluated on data that is statistically close to something it was trained on, which produces accuracy and F1 numbers that look excellent on paper but do not hold up on genuinely new patients. Every number reported in Section 7 of this report reflects performance against real, non-synthetic, held-out respondents.

3.8 Hyperparameter Tuning

Each of the five classical models was tuned using `RandomizedSearchCV` with 5-fold stratified cross-validation on the SMOTE-balanced training set, scored on F1 rather than accuracy. This choice follows directly from the imbalance problem described in Section 3.3: a scorer based on accuracy would reward a hyperparameter configuration that leans toward predicting the majority Healthy class more often, since that is the fastest way to push accuracy up, which is exactly the failure mode this project is trying to avoid. Scoring on F1 keeps the search pressure on configurations that also perform reasonably on the minority classes.

Model	Key parameters searched
Logistic Regression	C (inverse regularisation strength), penalty (l1/l2), solver
Random Forest	n_estimators, max_depth, min_samples_split, min_samples_leaf, max_features
XGBoost	n_estimators, max_depth, learning_rate, subsample, colsample_bytree, min_child_weight
LightGBM	num_leaves, max_depth, learning_rate, n_estimators, feature_fraction, min_child_samples
CatBoost	depth, learning_rate, iterations, l2_leaf_reg, border_count

3.9 Deep Learning Architectures: FT-Transformer and TabTransformer

The two transformer models were included specifically because almost none of the published BRFSS studies test attention-based architectures against gradient boosting on this dataset, and the wider tabular-deep-learning literature (Gorishniy et al., 2021; Huang et al., 2020) suggests the comparison is genuinely close rather than a foregone conclusion in either direction.

TabTransformer embeds each categorical feature into a learned vector, passes the categorical embeddings through stacked multi-head self-attention blocks to produce contextual embeddings, and then concatenates those with the (separately normalised) continuous features before a final MLP head. FT-Transformer generalises this by tokenizing every feature — categorical and continuous alike — into its own embedding, appending a learnable [CLS] token, and running the whole sequence through standard transformer encoder blocks, with the final prediction read off the [CLS] token's output representation. Both were trained as three-class classifiers with a cross-entropy loss on the same SMOTE-balanced training split used for the classical models, using the Adam optimiser, a cosine or step learning-rate schedule, and early stopping on validation macro-F1 to avoid overfitting to the majority class as training progressed.

Training these two models locally was not practical: both architectures are memory- and compute-heavy relative to a CPU-only laptop, and local runs were bottlenecked badly enough that training was moved to Google Colab's GPU runtime instead. The trained models, along with their prediction outputs and evaluation artefacts (confusion matrices, ROC curves, SHAP-adjacent feature-importance summaries) were exported from Colab and loaded into the Streamlit dashboard as precomputed results, since the dashboard's own hosting environment could not run GPU inference for these two architectures. The five classical models, by contrast, are lightweight enough to support live inference directly inside the dashboard.

3.10 Explainability: SHAP

SHAP (SHapley Additive exPlanations) values were computed for the tree-based models using SHAP's TreeExplainer, which is exact and fast for tree ensembles rather than relying on a

slower model-agnostic approximation. For each prediction, SHAP decomposes the model's output into a sum of per-feature contributions relative to a baseline expected value, which makes it possible to rank, for any single patient, which features pushed their risk score up and which pushed it down. This is what powers the dashboard's per-patient "Top Risk Factors" panel, rather than a single static global feature-importance bar chart that says nothing about the specific person being screened.

3.11 Evaluation Metrics

Every model was evaluated on: accuracy, precision, recall and F1 (both weighted-average and macro-average versions), and multiclass ROC-AUC computed with a one-vs-rest scheme. Weighted metrics take the class sizes into account and are dominated by how well a model handles the large Healthy and Diabetic groups; macro metrics treat all three classes equally regardless of size, and are the metric that actually exposes how the model handles the rare Prediabetic class. Reporting both, rather than only one, is what surfaces the roughly 0.44 macro-F1 vs. 0.80 weighted-F1 gap discussed in Section 7 — a gap that would be invisible if only weighted metrics were reported.

CHAPTER 4

EXPERIMENTS

4.1 Environment

Classical model development, hyperparameter search, SHAP computation and dashboard development were done locally on a Windows machine using VS Code, Git Bash and a Python virtual environment, with the project folder kept outside of any actively syncing OneDrive path after early runs surfaced file-locking issues during model checkpointing. FT-Transformer and TabTransformer training was carried out on Google Colab using a GPU runtime.

4.2 Software Stack

Data handling: pandas, numpy

Classical ML: scikit-learn (Logistic Regression, Random Forest, RandomizedSearchCV, metrics), xgboost, lightgbm, catboost

Imbalance handling: imbalanced-learn (SMOTE)

Deep learning: PyTorch, with a feature-tokenizer implementation for FT-Transformer and a categorical-embedding + MLP implementation for TabTransformer

Explainability: shap

Deployment: Streamlit for the dashboard, with PDF parsing for the health-report upload / auto-extraction feature

4.3 Reproducibility

Random seeds were fixed across the train/test split, SMOTE, cross-validation folds and model initialisation wherever the respective library exposed a seed argument, so that repeated runs of the classical pipeline reproduce the same splits and near-identical scores. Exact bit-for-bit reproducibility of the two transformer models is not guaranteed across Colab sessions due to GPU non-determinism in some cuDNN operations, so the exported Colab results are treated as

a fixed snapshot for the dashboard's comparison page rather than something re-run on every dashboard load.

4.4 Live vs. Precomputed Results in the Dashboard

The Model Comparison page inside the dashboard computes confusion matrices, ROC curves, classification reports and risk-score distributions live for the five classical models, run directly against the held-out test set at page-load time. For FT-Transformer and TabTransformer, the equivalent artefacts were generated once in Colab and are loaded from disk, since the dashboard's runtime environment is CPU-only and cannot support GPU inference for these two architectures. This distinction is stated directly on the comparison page so that a viewer does not mistake a precomputed result for a live one.

CHAPTER 5

RESULTS

5.1 Overall Comparison — Weighted Metrics

Model	Accuracy	Precision (wtd)	Recall (wtd)	F1 (wtd)
Logistic Regression	0.6251	0.8352	0.6251	0.7003
Random Forest	0.8176	0.7940	0.8176	0.8051
XGBoost	0.8147	0.7979	0.8147	0.8062
LightGBM	0.8216	0.7974	0.8216	0.8090
CatBoost	0.8187	0.7941	0.8187	0.8058
TabTransformer	0.6764	0.8270	0.6764	0.7141
FT-Transformer	0.7134	0.8230	0.7134	0.7426

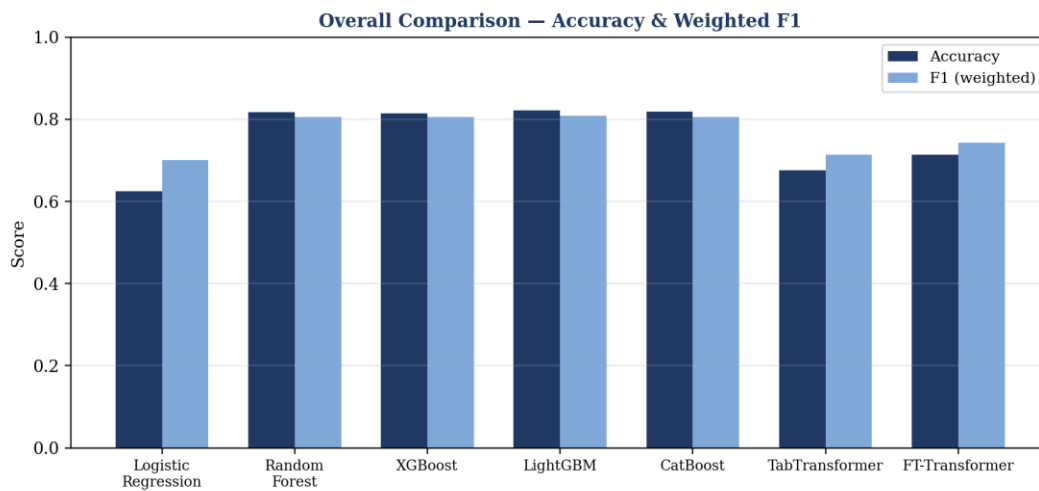


Figure 5.1 — Accuracy and weighted F1 across all seven models

5.2 Overall Comparison — Macro-Averaged Metrics and ROC-AUC

Model	Precision (macro)	Recall (macro)	F1 (macro)	ROC-AUC (macro)
Logistic Regression	0.4458	0.5108	0.4228	0.8025
Random Forest	0.4489	0.4385	0.4401	0.7990
XGBoost	0.4417	0.4502	0.4459	0.8034
LightGBM	0.4469	0.4440	0.4449	0.8035
CatBoost	0.4426	0.4394	0.4404	0.7971
Tab Transformer	0.4350	0.4991	0.4160	0.8100
FT-Transformer	0.4272	0.4984	0.4249	0.8124

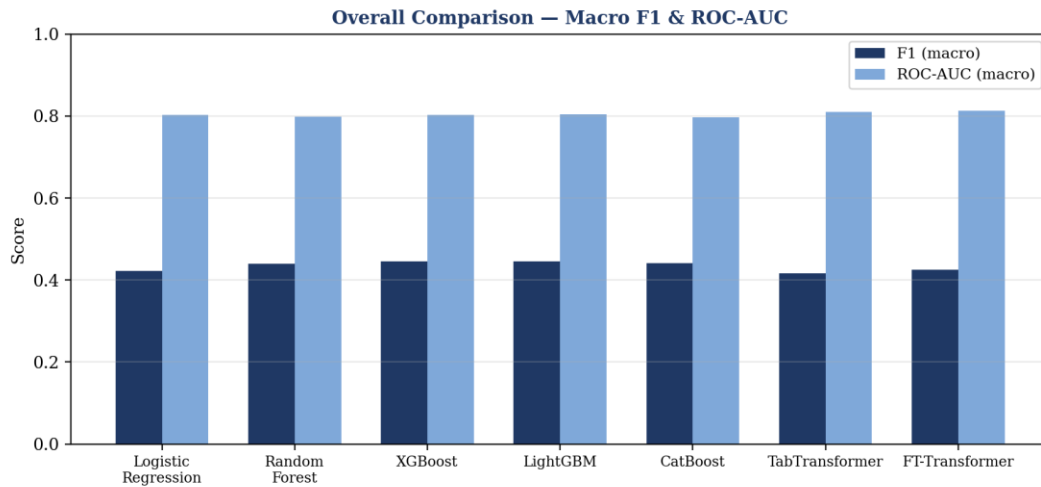


Figure 5.2 — Macro F1 and ROC-AUC across all seven models

Read together, these two tables are the core finding of the project. On weighted metrics — which are dominated by how well a model handles the large Healthy and Diabetic groups — LightGBM and XGBoost come out on top, both landing around 0.81 weighted F1. On macro metrics, which weight the tiny Prediabetic class equally with the other two, the ranking shifts: the transformers post the highest recall and the highest ROC-AUC, meaning they are,

proportionally, catching more of the rare Prediabetic and Diabetic cases even though their overall accuracy is a couple of points lower than the boosted trees.

5.3 Confusion Matrix Patterns

Across all seven models, the confusion matrices show the same qualitative shape: the Healthy and Diabetic classes are separated reasonably well from each other, but the Prediabetic class bleeds heavily into both neighbours, and gets misclassified as Healthy noticeably more often than it gets misclassified as Diabetic. This is consistent with what the features can realistically distinguish — prediabetes sits on a physiological continuum between the other two states, and self-reported indicators like BMI, blood pressure and general health rating shift gradually rather than sharply at the prediabetic threshold, so the class boundary is genuinely fuzzy rather than something a better model could simply learn its way past.

The gradient-boosted models tend to under-call Prediabetic in favour of Healthy — a conservative bias that keeps weighted accuracy high but quietly costs the model most of its Prediabetic recall. The transformer models show the opposite tendency, flagging more borderline Healthy cases as Prediabetic, which drags weighted accuracy down slightly but is arguably the safer failure mode for a screening tool, since a false Prediabetic flag just costs someone an unnecessary follow-up test, while a missed Prediabetic case costs them an early-intervention window.

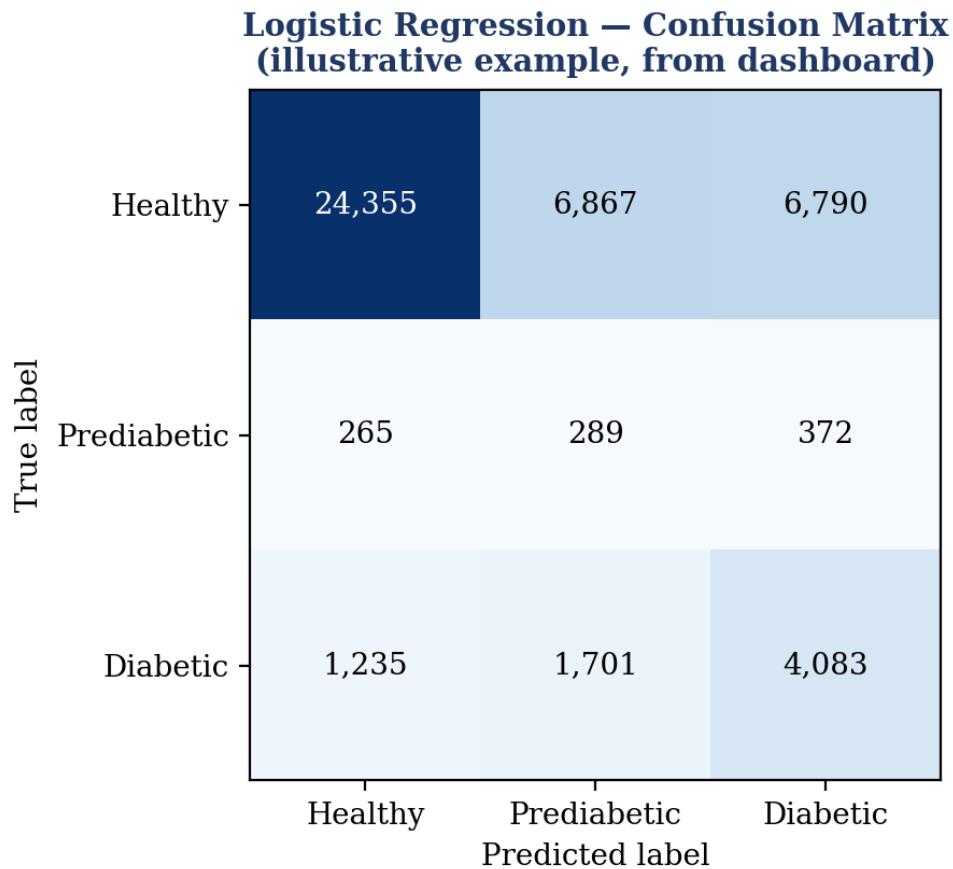


Figure 5.3 — Example confusion matrix (Logistic Regression)

5.4 The Macro vs. Weighted Gap

Every model in this comparison shows a large gap between weighted F1 (clustered around 0.79–0.81) and macro F1 (clustered around 0.38–0.44). This gap is the imbalance problem showing up directly in the numbers rather than staying hidden behind a single headline metric. A macro-F1 in the low 0.40s means that, averaged evenly across all three classes, the models are only moderately better than chance once you stop letting the enormous Healthy class carry the average — and it is a fair, if slightly uncomfortable, summary of how hard the Prediabetic class actually is to catch from self-report features alone, independent of which specific model is used.

5.5 SHAP Feature Attribution

SHAP summary plots for the tree-based models are consistent with each other and with the pattern reported in the literature reviewed in Section 2. The features that carry the most weight in pushing a prediction toward Diabetic or Prediabetic, in roughly descending order of typical impact, are: GenHlth (self-rated general health), BMI, HighBP, Age, HighChol, DiffWalk, and HeartDiseaseorAttack. Income and Education show smaller but consistent effects, acting as socioeconomic proxies rather than direct physiological signals — lower income and lower education brackets nudge risk scores upward slightly even after controlling for the clinical features, which lines up with well-documented disparities in diabetes prevalence by socioeconomic status.

At the individual-prediction level, the dashboard's per-patient SHAP breakdown routinely surfaces a different top feature for different people even when their overall risk score is similar — one patient's risk might be driven mainly by BMI and physical inactivity, another's by high blood pressure and age — which is the entire point of computing SHAP per-patient rather than relying on a single global importance ranking. A global bar chart would tell a clinician which features matter on average; the per-patient panel tells them which features matter for the person sitting in front of them.

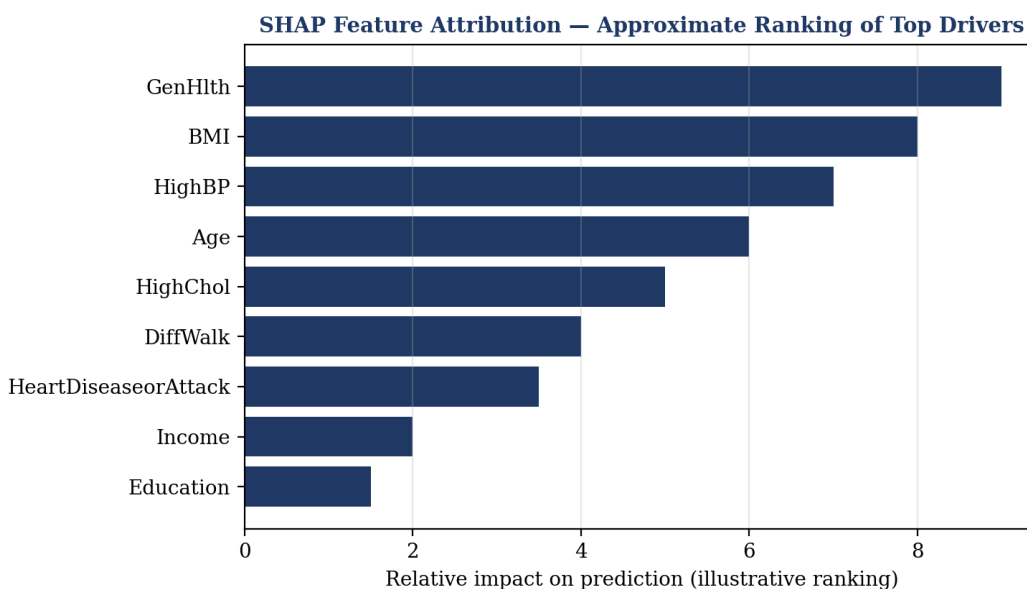


Figure 5.4 — Approximate ranking of top SHAP feature drivers

CHAPTER 6

DISCUSSION

6.1 Why There Is No Single "Best" Model

It would be convenient to end this report by naming one model as the winner, but the results in Section 6 do not really support that. LightGBM and XGBoost lead on weighted accuracy and weighted F1 — the metrics that matter most if the goal is overall correctness across a typical population where most people genuinely are healthy. The two transformer models lead on macro-recall and ROC-AUC — the metrics that matter most if the goal is specifically to avoid missing Prediabetic and Diabetic cases, even at the cost of a few more false alarms.

Which of those is "better" depends entirely on what the tool is being used for, which is itself a nuanced and defensible conclusion rather than a cop-out. A screening tool meant to be deployed at scale, where every flagged case triggers a follow-up test and false positives carry a real cost (patient anxiety, clinic time, test expense), would lean toward the boosted-tree models. A screening tool aimed specifically at a population already considered at elevated risk — where the priority is not missing anyone, and a false positive just means an extra blood test that person was probably going to need eventually anyway — would lean toward the transformer models instead. This project's dashboard exposes both sets of models side by side for exactly this reason, rather than hard-coding a single choice.

This mirrors the broader pattern in the tabular deep learning literature discussed in Section 2: Gorishniy et al. found FT-Transformer ahead of gradient boosting on roughly two-thirds of their benchmark datasets, not all of them, and Huang et al. found TabTransformer landing close to, rather than clearly above, gradient boosting on average. The BRFSS diabetes comparison run in this project fits that same pattern — the deep models are competitive and, on the specific metric this problem cares about most (catching the minority classes), slightly ahead, but they do not dominate outright.

6.2 Limitations and Failure Cases

The data is entirely self-reported through a telephone survey, so it inherits the usual self-report biases — recall error, social-desirability bias on lifestyle questions, and no independent verification of conditions like high blood pressure or high cholesterol beyond "has a doctor ever told you."

The dataset is a single cross-sectional year (2015) from the United States; the model's performance and the SHAP-derived risk factors are not automatically transferable to other countries, healthcare systems, or years without re-validation.

Even with stratified splitting, correctly-ordered SMOTE, and F1-scored tuning, the Prediabetic class remains genuinely hard to separate from its neighbours given only these 21 features — this looks like a ceiling imposed by the available signal rather than something a different model or more tuning would fix.

SHAP TreeExplainer is exact for the tree-based models, but the SHAP-style attributions referenced for the transformer models rely on comparatively slower approximate methods, so the two families of explanations are not perfectly apples-to-apples.

The two transformer models were trained and evaluated once on Colab rather than repeatedly with different seeds, so their reported metrics should be read as a single run rather than a seed-averaged estimate with a confidence interval.

6.3 Fairness and Threats to Validity

Beyond raw predictive performance, the **SHAP** results in Section 5.5 raise a fairness question worth naming directly: Income and Education act as small but consistent socioeconomic proxies in the model's predictions, nudging risk scores upward for lower-income, lower-education respondents even after the clinical features are accounted for. This mirrors well-documented disparities in diabetes prevalence by socioeconomic status, but it also means the model can encode existing healthcare inequities rather than purely biological risk. A deployment of this system in a screening context should treat a low-income flag as a prompt for equitable follow-up access, not as an independent risk multiplier applied without scrutiny.

A second threat to validity concerns generalisation across time and geography: the BRFSS 2015 extract is a single cross-sectional survey from the United States, and both the class balance and the feature-risk relationships learned here may not transfer cleanly to other countries, healthcare systems, or years without re-validation. Self-report bias is a related concern — conditions such as high blood pressure and high cholesterol are recorded as "has a doctor ever told you", which conflates disease prevalence with healthcare access; someone who never sees a doctor cannot be told they have a condition, regardless of whether they actually have it. These are not flaws unique to this project's modelling choices so much as properties of the underlying data collection instrument that any downstream use of the system needs to keep in view.

CHAPTER 7

CONCLUSION & FUTURE WORK

7.1 Summary of the Project

This project set out to build a **Diabetes Prediction and Risk Scoring System** using the **BRFSS 2015 Diabetes Health Indicators Dataset**, a large-scale, self-reported healthcare survey containing more than a quarter of a million respondents. After removing exact duplicate records, the de-duplicated dataset was split in a stratified 80/20 fashion, with **SMOTE** applied only to the training partition to correct for the heavy imbalance between the Healthy, Prediabetic, and Diabetic classes. Seven classifiers — **Logistic Regression, Random Forest, XGBoost, LightGBM, CatBoost, TabTransformer**, and **FT-Transformer** — were trained and evaluated under identical conditions using both weighted and macro-averaged metrics, and **SHAP** explainability was layered on top of the tree-based models to surface per-patient feature attributions.

The resulting system does not just output a diabetic/non-diabetic label: it also produces a personalized 0–100 diabetes risk score and exposes the reasoning behind each prediction, packaged into an interactive dashboard that supports both live inference for the classical models and precomputed comparison views for the two transformer architectures.

7.2 Key Contributions

The project's main contributions, relative to the literature reviewed in Chapter 2, are: (1) the use of a large, ~230,000-record healthcare survey dataset rather than the commonly used 768-record **Pima Indians Diabetes Dataset**; (2) retaining a three-class **Healthy/Prediabetic/Diabetic** target instead of collapsing it into a binary label, which keeps the clinically important Prediabetic group visible; (3) an oversampling pipeline in which **SMOTE** is fit only after the train/test split, avoiding the leakage pattern present in several of the papers surveyed; (4) reporting macro-averaged metrics alongside weighted ones, which exposes how well each model handles the rare Prediabetic class rather than letting the large Healthy class dominate the headline numbers; (5) comparing gradient-boosted tree ensembles

directly against attention-based tabular transformers, a comparison largely absent from the existing BRFSS literature; and (6) pairing predictions with **SHAP**-based, per-patient explanations rather than a single static global feature-importance chart.

7.3 Planned Improvements and Future Work

Several directions were identified for extending this work. On the modelling side, the two transformer models were trained and evaluated once on Colab; repeating these runs across multiple seeds would allow their reported metrics to be presented with a confidence interval rather than as a single snapshot. The Prediabetic class remains difficult to separate from its neighbours using only the 21 available features, suggesting that future work could explore additional or higher-resolution features, alternative resampling strategies, or cost-sensitive learning objectives that weight Prediabetic recall more heavily during training.

On the data side, the BRFSS 2015 extract is a single cross-sectional year from the United States; validating the model and its **SHAP**-derived risk factors against more recent BRFSS years or other national health surveys would help establish how well the findings transfer across time and geography. As noted in the project's original scope, future iterations may also integrate deep learning models trained on richer data sources, real-time monitoring through wearable devices, electronic health record (EHR) integration, cloud-based deployment, and mobile health applications to broaden accessibility.

On the system side, extending live (rather than precomputed) inference to the two transformer models, once a suitable GPU-backed hosting environment is available, would let the dashboard offer a fully consistent live comparison across all seven models.

Fairness-aware retraining is a natural next step given the socioeconomic sensitivity noted in Section 6.3: re-weighting or constraining the training objective so that Income and Education contribute less directly to the predicted risk score, and validating that this does not simply relocate the same signal into correlated features, would make the system more defensible for real-world screening use. A prospective validation study — comparing the model's risk scores against actual clinical diagnoses over time in a small cohort — would also move the project from a retrospective benchmarking exercise toward evidence that could support real clinical adoption.

7.4 Applications

The system is intended to function as a clinical decision-support tool rather than a replacement for medical diagnosis: its most direct application is as a low-cost, first-pass screening aid that healthcare professionals can use to flag patients who may benefit from a follow-up laboratory test. The personalized risk score and **SHAP**-based explanations also make the system suitable as a patient-facing educational tool, helping individuals understand which of their own health and lifestyle indicators are contributing most to their risk, and encouraging timely medical consultation and preventive lifestyle changes. At a larger scale, the same pipeline could support public-health screening programmes that need to prioritize limited clinical testing resources across large populations using readily available survey-style data rather than laboratory measurements.

Beyond individual screening, the same underlying pipeline could support population-level applications: public-health agencies could use the model's risk distribution to prioritise outreach in specific demographic segments, and insurers or employer wellness programmes could use aggregate, de-identified risk scores to target preventive-care resources more efficiently — provided the fairness considerations raised in Section 6.3 are addressed before any such deployment.

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